Meta^n: Recursive Self-Improvement through Emergent Depth

Zae Myung Kim · Young-Jun Lee · Seungyeon Jwa · Dongyeop Kang
University of Minnesota · Seoul National University · arXiv 2608.24735v1

HarnessRSI 67

evidence: retained

Paper

Code

1 What changes; what stays fixed

Retained artifact: solver wrappers and libraries.
Weights and the improver Ω never change.

1 Run the solver stack
Same backbone at solver and improver.
Collect scripts, scores and failure traces.

↓

2 Fixed Ω writes the next layer
Read traces + prior layer code.
Emit pre-process + callable helpers.

↓

3 Wrap; do not rewrite Ω
Pass strategic context inward;
prepend merged helpers to solver code.

↓

4 Check code; add child to archive
Invalid helper definitions are skipped.
A worse chain can still enter the archive.

↓

5 Select and repeat
Sample parents by score + exploration.
Keep per-task bests AND best single chain.

Stop: empty output, patience or finite caps.
Typical depth 3–4; deep specialists reach 6.

Designed recursion; adaptive depth and code. A depth-aware trace formatter partly scaffolds the reported layer roles (§2; App. D–E).

2 CO-Bench: archive ≠ one chain

GPT-5.2 · held-out test · 3 seeds · mean ± SD

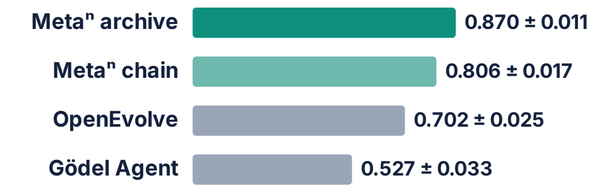


Table 2. Search archive uses per-task winning chains. Gemma: archive 0.851 ± 0.014, chain 0.782 ± 0.016, OpenEvolve 0.814 ± 0.022 — the single chain loses.

ARC dev, pass@2: archive 0.331 ± 0.010 vs GA 0.054 ± 0.006. No numeric held-out score located; qualitative wins only.

3 Controls and their boundaries

Gemma CO validation ablations · Table 3

Condition	Score
Full Meta^n	**0.845**
No library	0.825
No outer context	0.751
No recursion	0.714

Authors attribute ~72% of lift to context. Table has no row SD/CI. GPT depth-1 has no held-out pass or full-stack token-parity control.

- **OE-capped Gemma LawBench:** 0.784 ± 0.013 vs 0.745 ± 0.034. Full prompt runs spend 10–20× OE; CO described as near token parity.
- **GA corrected to per-task mode.** GPT s42 budget sweep reaches 0.628 at 11.4M tokens; Meta^n 0.870 is a 3-seed mean at stated 17M. Not exact parity.
- **Consolidation, 8 tasks / 3 seeds:** 0.71 ± 0.02 vs matched best-of-4 0.61 ± 0.03; gain CI [+0.04, +0.16]. Zero regressions via frozen non-focus traces.
- **Tuning:** 10-task CO pilot; default-selection split and baseline tuning-search budgets unspecified. No leakage audit reported.
- **Search cost:** Gemma CO s42 agentic 36.1M tokens; ARC ~$632 per seed. TB2: no OE/GA comparison; category SD, not seed SD.
- **Estimator caveat:** S2D held-out archive < chain on both backbones. Search-max definition versus test protocol unresolved (Tables 1–2).

4 Where depth fails to compound

- **Depth 3 interference:** mean lift −0.006; 41% of chain/task pairs regress. Mean by depth: 0.726 at 3 → 0.602 at 6 (§3.3).
- **S2D, GPT-5.2:** best chain 0.734 ± 0.020 vs OE 0.721 ± 0.007. Seed ranges overlap; no significance claimed.
- **AlgoTune, s42:** single-shot ×18.47 vs agentic ×15.96 on the same 7 tasks. Loss mostly from two FFT kernels; no per-task variance (Table 10).
- **AE Math:** GPT Meta^n/GA use 16 tasks, OE 10; not a common-task comparison. AE, SR and AlgoTune have no held-out split.

Takeaway

Richer context and per-task archive reuse help a fixed improver build better solvers. Recursion is designed; depth adapts but saturates. The archive can retain deep specialists without making every deeper chain better—or improving Ω itself.

Our read · HarnessRSI

Score 67 · two-star band · retained

Reports: held-out subsets, seeds, costs, ablations.
Unproven: improved Ω; transfer; universal parity.